

High-Sensitivity Vibration Monitoring of La Marota Viaduct Using a Novel 128-Channel Laser Radar System

Stephen Adrian O'Brien¹, Nicola Tarque², Jose Luis Rubio¹, Gregory Pandraud¹,
Eduardo Margallo¹, José Luis Martínez³ and José Maria Goicolea²

¹ Ommatidia LiDAR, Madrid, Spain

² Department of Continuum Mechanics and Structures, E.T.S.I Caminos, Canales y Puertos,
Universidad Politécnica de Madrid, Spain

³ INES Ingenieros Consultores, Madrid, Spain
stephen.obrien@ommatidia-lidar.com

Abstract: In this study, we employed a novel system combining massively parallel FMCW Laser Radar and Laser Doppler Vibrometry to monitor and analyze the structural vibrations of La Marota Viaduct, as part of a multi-year structural health monitoring project for high-speed railway infrastructure near Almodóvar del Río in southern Spain. The system features 128 parallel channels capable of non-contact velocity and displacement measurements with 40kHz sampling rate, allowing the real-time capture of dynamic structural behavior across a wide range of frequencies from a single installation point. The optical measurements were in good agreement with the numerical models and the accelerometer and strain gauge results. The exceptional sensitivity of the interferometry-based technique, particularly in low-frequency domains below 2 Hz, combined with the large number of measurement points offers the potential for superior model performance compared to traditional accelerometer setups. The high resolution and rapid acquisition capabilities of the system facilitated detailed analysis of the bridge's vibrational modes, providing insights into structural health and potential areas of concern.

Keywords: Laser RADAR, vibrometry, SHM, bridge monitoring.

1 Q1 Laser Radar

1.1 Ommatidia Laser Radar 's solution

The system, with 128 simultaneous measurement channels, employs Frequency Modulated Continuous Wave (FMCW) Laser Radar [1] and Laser Doppler Vibrometry (LDV) [2] techniques for ranging and velocity measurement, respectively.

In the FMCW approach, a continuous-wave laser undergoes frequency modulation, and the ranging information is extracted through coherent detection by analyzing the beating signal generated from the interference between the local oscillator and the return signal. The use of interference based coherent detection makes the system also sensitive to the Doppler-shift effect of the reflected light,

enabling the use of the LDV approach to extract the velocity or vibration information via coherent detection of the scattered light and the reference beam. This approach to optical offers a number of advantages:

- Enhanced Signal-to-Noise Ratio (SNR) through coherent detection.
- Continuous-wave laser operation, eliminating the need for high peak power conditions.
- Intrinsic Doppler velocity measurement, enabling simultaneous distance and velocity determination.
- Full complex signal acquisition, which allows for advanced signal processing.

A key enabling innovation in this system is the integration of Ommatidia LiDAR's Lightfield concept, leveraging the full complex return signal to implement a beam-forming network in the receiver. This enables lens-less lateral resolution, enhancing spatial imaging capabilities, and allowing the parallel detection of the aforementioned 128 simultaneous channels. More details on the Lightfield concept combined with FMCW can be found in [3]. The full description of the instrument placement and location is given in the following section.

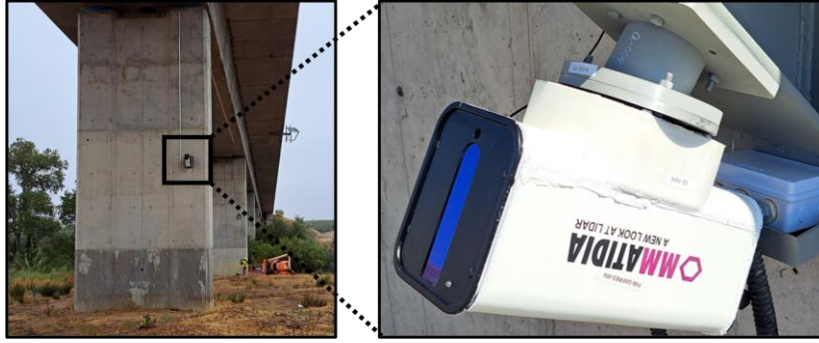


Fig. 1. Q1S system used in this work, as installed in the configuration of bridge monitoring.

1.2 System architecture

The two main components of the system are a light field sensor that receives the light reflected from the scene, and a tunable light source (DFB laser), which acts as the emitter. Additionally, the light is collected and emitted through dedicated optics. On the Optical Scanning Plane (OSP), 128 optical channels are collected simultaneously using the beamforming network of the Photonic Integrated Circuit (PIC) Receiver, while illuminating the scene with a line of light matching the angular field of view of the receiver. The proposed imager processes the electrical outputs to resolve the relative phase and power of the incident wave at each pixel. The photons are then converted into electrical signals and read out. Algorithms allow the generation of point clouds. The FMCW concept is applied here, building on the advantages of this technique (heterodyne gain, coherence gating, coherent detection) while eliminating the need for scanning and the use of collimated beams.

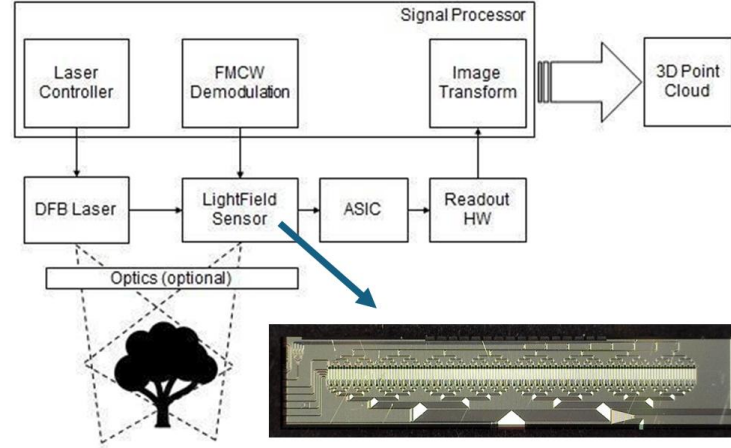


Fig.2. Overall system architecture with a zoom in of the PIC receiver at the core of the Laser Radar.

The PIC Receiver consists of an integrated Photonic Integrated Circuit, based on Silicon Nitride [4] technology. 128 optical channels collecting the return radiation are mixed in phase and in quadrature with the local oscillator coming from the illumination path. The beating signal is detected by external detectors, and the analog signal is read out by dedicated electronics. The latter is further interfaced with an FPGA, which processes the data stream to provide the 4D point clouds and Doppler-based instantaneous velocity measurements (Fig.2).

2 Implementation at the La Marota Viaduct

Ommatidia LiDAR has successfully installed four units of its Q1 system on a high-speed train bridge, with each unit covering a separate span of the structure as shown in Figure 4. The La Marota Viaduct carries the Córdoba-Málaga high-speed rail line across the shallow valley of the Arroyo de La Marota in Spain.

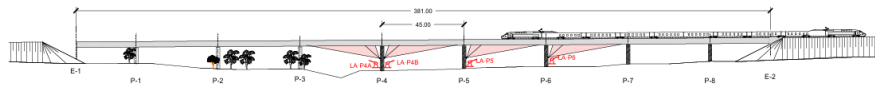


Fig. 3. Position of the four Q1s Laser Radar and LDV systems installed under the La Marota Viaduct.

The bridge is a continuous concrete box girder, 14 m wide, that accommodates two tracks. It has a total length of 380 m. With a depth-to-span ratio of 1:14, it is a typical stiff high-speed railway viaduct (Fig.3). The viaduct is supported by two abutments and eight intermediate rectangular piers measuring 5.5 x 2 meters, with heights varying from 10 to 15 meters. This configuration results in seven central spans of

45 meters each and two end spans of 33 meters. All supports, using pot-type constrained elastomeric bearings, accommodate longitudinal movement and bending about both vertical and transverse axes. One abutment is fixed longitudinally, while the other allows free movement. The viaduct's cross-section is a single-cell box girder with cantilevered top slabs. Five distinct cross-sectional types are incorporated along the viaduct's length, all maintaining a consistent deck width of 14 meters, sufficient to carry two standard-gauge railway tracks on ballast (Fig.4).



Fig. 4. Views (outside and inside) and plan view of the La Marota viaduct (left). The nomenclature for the end abutments, internal piers and spans are shown in the plan view (right).

Each span of the bridge is monitored by 128 speed signals (the optical channels described in section 1.2), uniformly spaced at 0.49° across a 62° Field of View (FoV). This specific FoV has been chosen to collect information over the full span. The LiDAR is also positioned at an angle providing a large density of points at key locations such as at the interface between the supporting pillar and the bridge deck. To enhance sensitivity, a retroreflective paint was applied, ensuring precise and accurate data collection. A careful alignment of the 128 laser beams onto the painted line is done for all units, ensuring optimal operating conditions (Fig.5).

Each of the Q1 systems are powered through a single Power over Ethernet cable. Similarly data connection is done using this cable Fig.5.

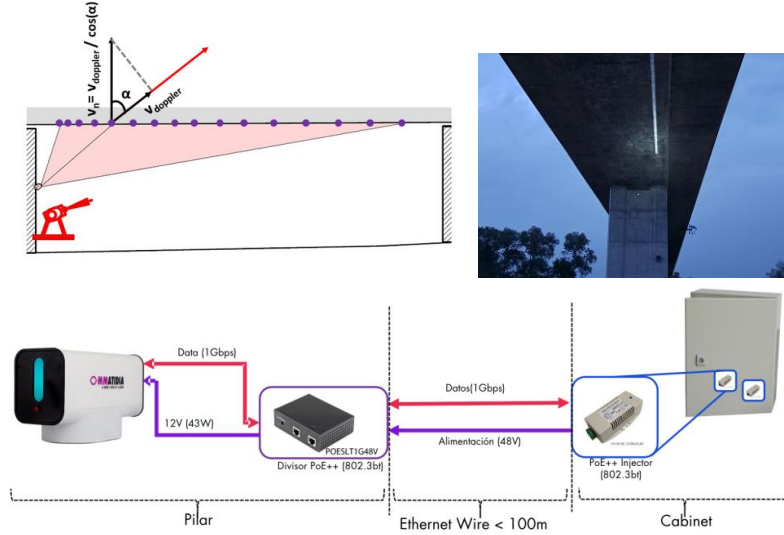


Fig. 5 Top left: Laser Points distribution under the bridge offering a full coverage of the bridge Top right: A painted line improves the reflection of the light and the overall system performances. Bottom: Power and data collection for the Q1S units.

3 A 3D beam numerical model

A finite element (FE) model of the La Marota Viaduct was previously [5] developed to analyse its dynamic behaviour. While a detailed 3D shell element model was initially considered for its accuracy in representing geometry, boundary conditions, and local deformations, its high computational cost led to adopting a simplified 3D beam element model for this study.

This beam FE model was implemented in Abaqus, using 3D beam-column elements. The nine-span hyperstatic structure (seven 45m spans and two 33m spans) was modelled, with each span incorporating five distinct caisson cross-sections. These sections were defined as general sections, combining their respective geometries and calibrated mechanical properties. Non-structural mass was also included, accounting for dead loads such as ballast, rails, sleepers, and guard walls.

Boundary conditions (BCs) were applied, reflecting design specifications and validated through field observations. At all supports (two abutments and eight piers), transverse (Z-axis) and vertical (Y-axis) displacements, as well as rotation about the longitudinal axis (X-axis), were constrained. At the fixed abutment (E-1, Figure 7a), longitudinal displacement and Y-rotation were also restricted, but at the initial centroid of the beam section to accurately replicate the connection detail. The partial longitudinal constraint at abutment E-1, provided by pot-type elastomeric bearings, was modelled using a spring element. Multi-point constraints (MPCs) were used to connect the BCs at the shifted longitudinal axis to the original centroid

axis. Similarly, the partial rotational constraint provided by the elastomeric pot bearings at the piers and right abutment was simulated using rotational spring elements (Fig. 6 b).



Fig. 6 a) Internal view of the fixed abutment E-1, b) View of an elastomeric pot.

Fig. 7 a) shows the 3D beam finite element model of the La Marota viaduct constructed in Abaqus. Fig. 7 b) shows the viaduct's deformation under a simulated railway load and below the history deflection at the central point of the 7th span. The calibrated eigenvalues of the digital twin model were compared with the experimental ones obtained through an OMA, and the results are shown in Table 1 for the first five frequencies.

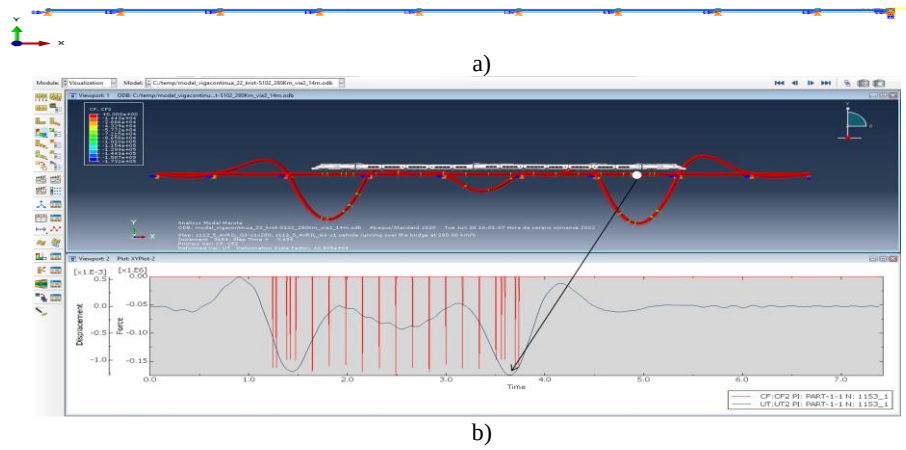


Fig. 7. a) 3D beam beam finite element model in Abaqus, b) example of the deformation under the pass of a train.

Table 1. Calibrated eigenvalues vs experimental ones.

(Hz)	f_1	f_2	f_3	f_4	f_5
Experimental	3.08	3.87	4.50	5.74	6.08
FEM	3.03	3.67	4.69	5.72	6.97
Variation %	1.42	5.02	-4.13	0.40	14.66

4 Laser Doppler Vibrometry Measurements

The flexural modes of interest range from 3.0 to 9 Hz. The trains operate at speeds of up to 300 km/h. A complementary monitoring system features accelerometers at the center of each span (PBC 393A03) and strain gauges on the track rails, allowing for the measurement of train velocities and axle loads. Peak accelerations during train passages can reach up to 2 m/s^2 .

4.1 Comparison with accelerometers

The comparison is made at mid-span of the 7th span, with the accelerometer placed under the South track (Málaga -> Córdoba direction).

The LDV data was obtained with the Q1S system installed at the 6th pillar and pointing towards the 7th span. The data was acquired while a train was passing over the track, under which both the accelerometer and the Q1S LDV measuring points are located.

In Fig. 8 (left) we show a plot of the acceleration reading obtained directly from the A03 vs. the derived velocity measurement from the Q1S by the point located at the same position. On the right-top side we show a detail of the signal *tails* after the train has exited the span, and on the right-bottom side we show the frequency spectrum of the central part of the signal.

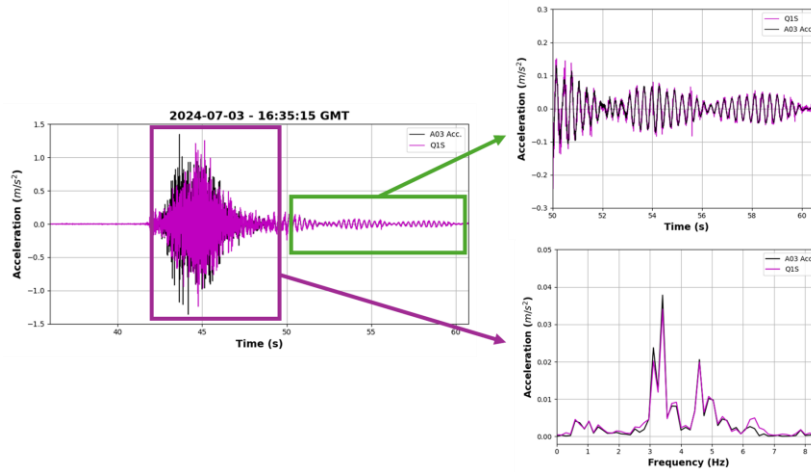


Fig. 8. Left) Comparison of A03 acceleration and Q1S velocity derivative measurements. Right top) Detail on the signal *tails* Right bottom) Frequency spectrum of the central part of the signal

4.2 Numerical vs Experimental VVI comparison

This section details the multichannel laser vibrometry measurements and their interpretation using the velocity vibration index (VVI). These experimental VVI

measurements are then compared with corresponding results from the numerical model (digital twin). The VVI, a measure of the spatial extent of vibration, is calculated using vibration velocity data acquired over the central 25 meters of spans 4, 5, 6, and 7 of the La Marota Viaduct (see Fig. 3).

Specifically, VVIs were calculated from measurements taken by 31 channels, ranging from channel 70 (at 10.01 m) to channel 100 (at 16.97 m), see Fig. 9. The following expression defines the VVI:

$$VVI = \frac{1}{N} \frac{1}{L} \sum_{j=1}^N \sum_{i=1}^k (l_i v_i^2)_j$$

with N the time records in the episode of duration T , l_i the length of the tributary board of each point, L the total length of the area under analysis and α_i the angle of each ray with the vertical.

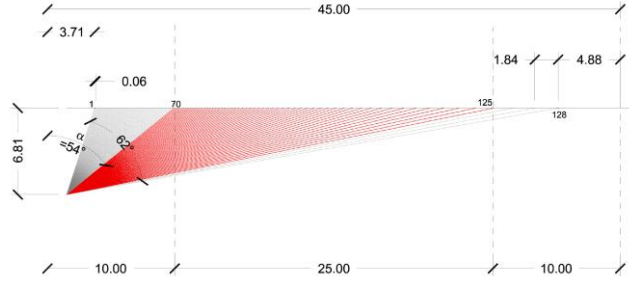


Fig. 9. Central area of laser incidence on a span of the structure.

One case study is analysed to validate the VVI under real train loads. The train travelled in the Córdoba-Madrid direction. Measurements recorded by lasers LA-P5 and LA-P6 on spans 6 and 7, (respectively Fig. 10 and 11), were compared with results from the digital twin model (filtered up to 10 Hz). The first case involved a double-composition S112 train passing at 260 km/h. As shown in Table 2, the error between the numerical and experimental VVI values is less than 5%, which is considered acceptable given the high sensitivity of the VVI calculation.

Table 2. Comparison of the experimental and numerical VVI for one case studies (Train S112 double at 260 km/h).

Span	6	7
Experimental	1.92E-06	1.94E-06
Numerical	1.99E-06	2.03E-06
Difference	3%	5%

A comparison of the velocity time-history plots of the experimental and numerical measures for the case study is shown in the figure below:

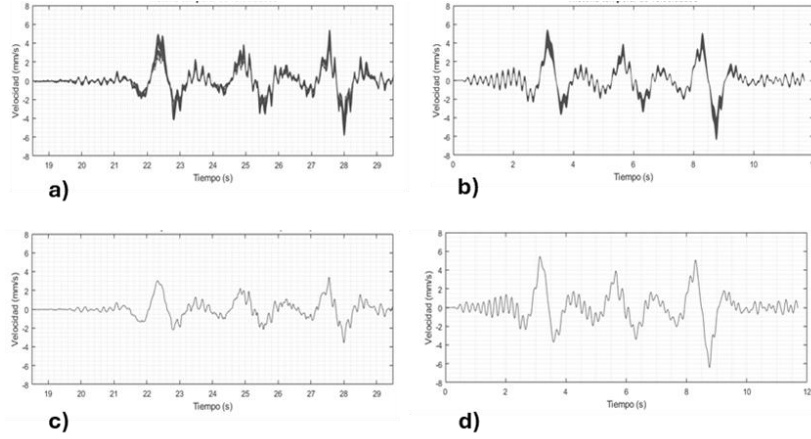


Fig. 10. Pillar 5 (6th Span) Train S112 double, vel. 260 km/h: a,c) Experimental, all channels (a) and channel 70 (c) velocity time-history; b,d) FEM, all channels (b) and channel 70 (d) velocity time-history.

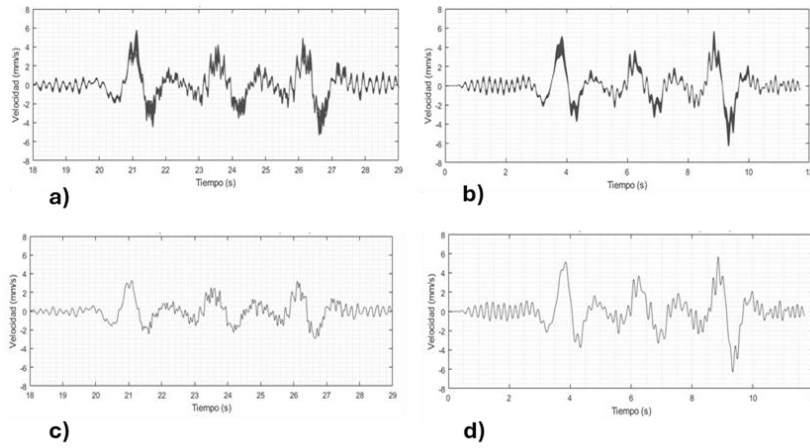


Fig. 11. Pillar 6 (7th Span) Train S112 double, vel. 260 km/h: a,c) Experimental, all channels (a) and channel 70 (c) velocity time-history; b,d) FEM, all channels (b) and channel 70 (d) velocity time-history.

4.3 Numerical vs Experimental relative displacement comparison

Finally we have also analysed the relative displacement values at mid-span obtained by integrating the LDV data, and compared it with the numerical results. Since the Q1S velocity information provides good information even at very low frequencies, the velocity information can be integrated with high accuracy. Fig. 12 shows an example of this comparison, showing the excellent agreement between both of them.

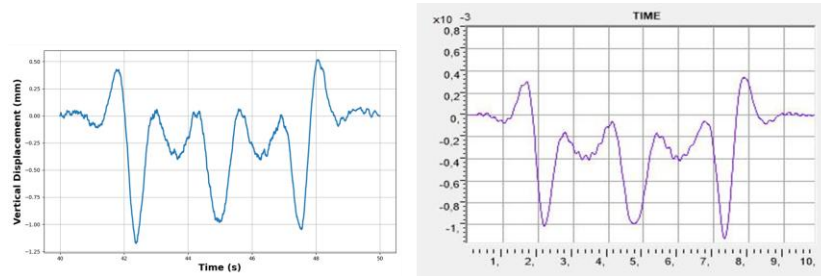


Fig. 12. Relative displacement at the middle of the 7th span obtained from the velocity information measured by the Q1S (left) and simulated with the digital twin model (right).

5 Conclusions

In conclusion, the study demonstrates that the novel 128-channel Laser Radar system, leveraging both FMCW and LDV techniques, provides highly sensitive, real-time monitoring of structural vibrations in the La Marota Viaduct. The system's ability to capture sub-millimeter displacements and accurately resolve low-frequency vibrational modes significantly enhances the detection and analysis of dynamic structural behaviors, offering a promising alternative to traditional accelerometer-based methods.

The strong agreement between optical measurements and both numerical predictions and accelerometer measurements validates the system's effectiveness, underscoring its potential to advance structural health monitoring and predictive maintenance strategies for critical infrastructure.

References

1. Martin A et al., Photonic Integrated Circuit-Based FMCW Coherent LiDAR, in Journal of Lightwave Technology 36(19): 4640-4645.
2. Rothberg SJ et al., An international review of laser Doppler vibrometry: Making light work of vibration measurement, in Optics and Lasers in Engineering 99, 11-22.
3. Guivernau JL, Gonzalo IB, Margallo-Balbás E, and Pandraud G, (2021) High-Performance Akinetic Parallel Light Field Sensor for 3D Imaging. In: Conference on Lasers and Electro-Optics, J. Kang, S. Tomasulo, I. Ilev, D. Müller, N. Litchinitser, S. Polyakov, V. Podolskiy, J. Nunn, C. Dorrer, T. Fortier, Q. Gan, and C. Saraceno, eds., OSA Technical Digest, paper ATu4K.7.
4. Shaw M, Guo J, Vawter G, Habermehl S and Sullivan, C, (2005). Fabrication techniques for low-loss silicon nitride waveguides. Proceedings of SPIE - The International Society for Optical Engineering. 5720.
5. Tarque, N, Goicolea JM, Cuevas RG, and Martínez JL. (2024) Finite element model updating for the dynamic response of continuous deck railway viaducts leading to efficient digital twins in *XII International Conference on Structural Dynamics*, Journal of Physics: Conference Series, vol. 2647, p. 102009.